ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, examples and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 A ✓ 1.1.3 C ✓ 1.1.4 C ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 D ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 B ✓ 1.3.5 E ✓ 1.3.6 H ✓ 1.3.7 J ✓ 1.3.8 F ✓ 1.3.9 C ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A robot may not injure a human being or, through inaction, allow a human to come to harm ✓. A robot must obey orders given by humans, except where they conflict with the first law ✓. A robot must protect its own existence as long as this does not conflict with the first or second law ✓. (3)

2.2 Any THREE classes ✓✓✓ each with an example ✓✓✓: industrial (welding / painting / assembly arm); service (cleaning, delivery, hotel or restaurant robot); medical (surgical robot, rehabilitation); field (mining, agriculture, underwater, space rover); military / security (bomb disposal, patrol); educational / entertainment (toy, school kit). (6)

2.3 Any FOUR ✓✓✓✓: safety fences / guards around the cell; interlocked gates that stop the robot when opened; emergency-stop buttons; light curtains or safety mats; warning lights and signs; reduced speed when people are near; training and lock-out procedures; marking the work envelope on the floor. (4)

2.4 Any ONE ✓: job losses for low-skilled workers; who is responsible when a robot causes harm; privacy of data collected by robots; cost and access (only rich companies benefit); the need to re-skill workers. (1)

[14]

QUESTION 3

3.1 Any FOUR named and described ✓✓✓✓: manipulator / arm – links and joints that move the tool; end effector – gripper or tool that does the work; controller – the computer that stores the program and controls the motion; drives / actuators – motors, hydraulic or pneumatic units that move the joints; sensors – give feedback on position and the surroundings; power supply; teach pendant. (4)

3.2 6 DOF ✓: three joints in the arm (waist, shoulder, elbow) plus three wrist movements (pitch, yaw, roll) ✓ (2)

3.3 Accuracy: how close the robot gets to the exact commanded position ✓✓. Repeatability: how close it returns to the same position each time it repeats the move ✓. Example: a robot that always lands 2 mm to the left of the target is repeatable but not accurate ✓. (4)

3.4 Any TWO grippers ✓✓ each with a suitable object ✓✓: two-finger mechanical gripper – blocks, boxes, parts; vacuum / suction cup – flat sheets, glass, cardboard boxes; magnetic gripper – steel sheets and parts; soft / adaptive gripper – fruit, eggs, delicate items; hooks – bags, crates. (4)

3.5 $\mathrm{F} = (\mathrm{m} \times \mathrm{g}) \div (\mu \times \mathrm{n})$ ✓ $= (2 \times 9{,}8) \div (0{,}4 \times 2)$ ✓ $= 24{,}5$ N ✓; with safety factor: $24{,}5 \times 2 = 49$ N per finger ✓ (4)

[18]

QUESTION 4

4.1 Open loop: no feedback, the controller sends a command and does not check the result ✓, e.g. a stepper motor driven a fixed number of steps / a timer-controlled conveyor ✓. Closed loop: the output is measured and compared with the set point, and the error is corrected ✓, e.g. a servo joint with an encoder / a line-following robot ✓. (4)

4.2 Blocks ✓✓✓✓ and correct loop with arrows ✓: set point (desired angle) → comparator / summing point (error) → controller → motor driver / actuator → robot joint (actual angle); sensor (encoder / potentiometer) feeding the actual angle back to the comparator. (5)

4.3.1 $\theta = 360° \div 600 = 0{,}6°$ ✓ (1)

4.3.2 $\mathrm{C} = \pi \times \mathrm{d}$ ✓ $= \pi \times 65 = 204{,}20$ mm ✓ (2)

4.3.3 $\text{Distance per pulse} = 204{,}20 \div 600$ ✓ $= 0{,}34$ mm ✓ (2)

4.3.4 $\text{Revolutions} = 1\,000 \div 204{,}20 = 4{,}90$ ✓; $\text{pulses} = 4{,}90 \times 600 \approx 2\,938$ pulses ✓ (accept 2 938–2 941) (2)

4.4 P: gives a correction proportional to the size of the error (bigger error, bigger push) ✓. D: reacts to how fast the error is changing, damping the motion and reducing overshoot ✓. (2)

[18]

QUESTION 5

5.1 $d = (v \times t) \div 2$ ✓ $= (343 \times 0{,}0029) \div 2$ ✓ $= 0{,}50$ m ✓ (3)

5.2 Any ONE feature ✓ and ONE use ✓ for EACH: DC motor – simple, fast continuous rotation, speed set by voltage / PWM – drive wheels; stepper – moves in exact steps without feedback, holds position – 3D printer axes, positioning tables; servo – turns to a commanded angle using built-in feedback – robot arm joints, grippers, steering. (6)

5.3 Lead-through / walk-through ✓ – easy, operator simply guides the arm ✓; teach pendant ✓ – precise points can be stored and edited on the shop floor ✓; offline (simulation / text) programming ✓ – done on a computer without stopping production, can be tested in a simulator ✓. (6)

5.4 Internal sensors measure the robot's own state ✓, e.g. encoder, joint potentiometer, battery-level sensor, IMU ✓. External sensors measure the surroundings ✓, e.g. ultrasonic, camera, LiDAR, touch / bump sensor ✓. (4)

5.5 LiDAR ✓ (1)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE